

REVIEW

Gamut of Patients Referred to Cardiology for Question of Clonal Hematopoiesis

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ABSTRACT: Clonal hematopoiesis encompasses diverse somatic mutations in hematopoietic cells, ranging from age-related expansions to mutations acquired after cytotoxic therapy, with implications for hematologic malignancy and cardiovascular disease. We characterize the spectrum of patients referred to cardiology for clonal hematopoiesis, including incidental detection during cytopenia or cancer predisposition workup, coexisting malignancy, and posttherapy surveillance. High-risk features, large clone size, multiple mutations, and specific driver genes interact with traditional cardiovascular risk factors to influence ischemic events. Contextualizing clonal hematopoiesis by detection setting, genotype, and clinical history informs individualized cardiovascular evaluation and risk mitigation, guiding mechanistically targeted preventive strategies and trial design.

Key Words: cardiology ■ cardiovascular diseases ■ clonal hematopoiesis ■ cytopenia ■ heart disease risk factors ■ hematologic neoplasms ■ mutation

Clonal hematopoiesis (CH) represents a heterogeneous state of clonal expansion within the blood-forming compartment. CH may persist for years without clinical manifestation or may progress to hematologic malignancy. In some individuals, hematopoietic stem cells acquire somatic mutations in a subset of leukemia driver genes, enabling positive selection and contributing to the risk of progression to myeloid or lymphoid malignancies, a phenomenon known as CH of indeterminate potential (CHIP).¹ If cytopenia were to exist in an individual with CH, the condition would be classified as clonal cytopenia of unknown significance (CCUS), which differs from myelodysplastic syndrome due to the absence of morphological dysplasia.² While CHIP and CCUS have a well-established link to hematologic malignancies, CHIP can entail an elevated risk of cardiovascular disease, including a 2-fold increased risk of incident coronary heart disease or ischemic stroke.³

For clarity, CHIP is defined as the presence of a myeloid-associated somatic mutation with variant allele frequency (VAF) $\geq 2\%$, a threshold used in most population-based studies linking CH to cardiovascular outcomes. A threshold of VAF $\geq 10\%$ has been denoted large CHIP in many studies. Many clinical and research

sequencing platforms can detect smaller clones; however, the clinical significance of these continues to be investigated. Risk models may also use different VAF cut points for specific end points: for example, the CH risk score incorporated maximum VAF $\geq 2\%$ as one predictor of myeloid neoplasm development among individuals meeting CHIP/CCUS criteria,⁴ whereas real-world CCUS cohorts have identified higher VAF thresholds (eg, 22%) that best stratified leukemia-free survival.⁵ Here, we refer to large CH as VAF $> 10\%$, which consistently associates with higher hematologic progression risk and cardiovascular disease risk across cohorts, while recognizing that the most discriminative VAF threshold may vary by clinical context, assay, and chosen outcome.

As experimental and epidemiological work continues to unravel the biological mechanisms and cellular implications of CH, the diverse and complex effects of different somatic mutations have become increasingly evident. We present here the spectrum of patients referred to the cardiovascular specialist for the question of CH. Through case-based discussion, we outline 4 distinct clinical categories of CH and propose a structured approach to their individualized cardiovascular evaluation and management.

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Nonstandard Abbreviations and Acronyms	
CCUS	clonal cytopenia of unknown significance
CH	clonal hematopoiesis
CHIP	clonal hematopoiesis of indeterminate potential
DDR	DNA damage response
HDL-C	high-density lipoprotein cholesterol
LDL-C	low-density lipoprotein cholesterol
Lp(a)	lipoprotein(a)
VAF	variant allele frequency

CH DETECTED DURING WORKUP FOR BLOOD COUNT ABNORMALITY

Case 1

An 81-year-old man with medical history notable for hypertension, dyslipidemia, past cigarette smoking, rheumatoid arthritis, and coronary artery disease was found to have CH while undergoing an evaluation for mild anemia. Eighteen months before referral, the patient developed exertional chest discomfort due to unstable angina and underwent percutaneous coronary revascularization of the mid-left anterior descending artery and proximal right coronary artery, as well as balloon angioplasty of the first diagonal branch and right posterior descending artery. The patient's chest discomfort resolved, but he subsequently developed fatigue and was found to have mild anemia and thrombocytopenia (hemoglobin, 12.5 g/dL; platelet, 144 K/uL). Next-generation sequencing (NEBNext kit, New England Biolabs) demonstrated mutations in the *ASXL1* (13.6% VAF) and *TET2* (10.7% VAF) genes. The patient's regular medications included aspirin 81 mg daily, amlodipine 5 mg daily, losartan 100 mg daily, and atorvastatin 40 mg daily. On examination, his heart rate was 70 bpm and blood pressure was 120/80 mm Hg, with a body mass index of 24.8 kg/m². Laboratory examination showed hemoglobin of 11.9 g/dL, leukocytes of 4.0 k/ μ L, platelets of 153 k/ μ L, total cholesterol of 162 mg/dL, triglycerides of 109 mg/dL, LDL-C (low-density lipoprotein cholesterol) of 88 mg/dL, HDL-C (high-density lipoprotein cholesterol) of 52 mg/dL, and HgA1C of 5.7%.

CCUS is defined as the presence of somatic mutations belonging to the myeloid gene panel with concomitant persistent unexplained cytopenia that does not fulfill diagnostic criteria for a hematologic neoplasm. Similar to CHIP, CCUS is a precursor for myeloid neoplasms and has a distinct genetic and cytogenetic profile.⁶ The community-level prevalence of CCUS remains unknown owing to retrospective study designs, small cohort size,

and referral bias. However, insights about the prevalence of CCUS in patients undergoing cytopenia workup have emerged from a British prospective cohort study. Of adult patients referred for investigation of cytopenia, 27% had CCUS.⁷ Compared with individuals with idiopathic cytopenia, those with CCUS were older and were more likely to be male. The most frequently mutated genes in individuals with CCUS were *TET2*, *SRSF2*, and *DNMT3A*.⁷ The natural history of individuals with CCUS is heterogeneous and depends on clonality and cytopenia patterns. A model developed to predict the risk of myeloid neoplasms identified the presence of splicing variants, platelet count, and number of somatic mutations as key prognosticators enabling the creation of a multiparametric risk score.⁵

Most contemporary risk prediction tools in CCUS do not provide information on the incidence of cardiovascular disease. However, the CH risk score, a prognostic formula derived from demographic, complete blood cell count, and molecular factors, correlates with cardiovascular disease mortality (Table).⁸ Although originally developed to stratify the risk of individuals with CHIP-CCUS for myeloid malignancy, the CH risk score may help identify individuals with increased susceptibility to ischemic cardiovascular events. In a population-based prospective cohort study, cardiovascular disease accounted for the most common cause of mortality in individuals classified by the CH risk score as having high-risk CHIP-CCUS ($\approx$ 21% versus 7.1% in those at low risk).⁸ In addition, the standardized hazard ratio of cardiovascular death in individuals with high-risk CHIP-CCUS was 2.9 compared with those with no CH (low risk; standardized hazard ratio, 1.03).⁸

Case Continued

In addition to the *TET2* mutation, this patient harbored a large *ASXL1* clone. *ASXL1* mutation increases the risk of myeloid malignancy. While its cardiovascular implications are less well mechanistically defined than those of *TET2*

Table. CHRS Parameters (4)

Parameter	Description
Single <i>DNMT3A</i>	Lower risk if present
High-risk mutation	Splicing factor genes, AML-like genes, <i>JAK2</i> , and <i>TP53</i>
Mutation number	Higher risk if ≥ 2
Variant allele fraction	Higher risk if ≥ 0.2
Red cell distribution width	Higher risk if ≥ 15
Mean corpuscular volume	Higher risk if ≥ 100
Cytopenia	Lower risk with CHIP than CCUS (for hematologic events)
Age, y	Higher risk if ≥ 65

CCUS indicates clonal cytopenia of unknown significance; CHIP, clonal hematopoiesis of indeterminate potential; and CHRS, clonal hematopoiesis risk score.

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or *JAK2*, *ASXL1*-mutant CH has been linked to systemic inflammation and accelerated atherogenesis through a mechanism involving Interleukin-1 Receptor–Associated Kinase 1 signaling.⁹ CH due to *ASXL1* mutations may contribute to vascular risk, particularly in the setting of large clone size and coexisting cardiovascular risk factors, but additional work is needed to establish this association definitively. Given the patient's coronary artery disease in the presence of multiple high-risk enhancers, including high VAF *TET2* CCUS and rheumatoid arthritis, LDL-C target <55 mg/dL was selected, applying a lipid-lowering target analogous to that recommended for very-high-risk populations. We recommended the addition of ezetimibe 10 mg daily to his high-intensity statin. Vitamin C supplementation was discussed based on pre-clinical data demonstrating partial restoration of *TET2*-dependent DNA demethylation¹⁰; however, no clinical trial evidence currently supports cardiovascular benefit in *TET2*-CH.

CH DETECTED DURING CANCER PREDISPOSITION WORKUP

Case 2

A 35-year-old woman with medical history notable for hypertension, obesity, and dyslipidemia was referred to us after an incidental finding of *NF1* CH detected on a multicancer gene panel to evaluate for germline predisposition, given the family history of breast cancer. The test was positive for a pathogenic variant in the *NF1* gene (c.1949_1967dup; p.Ser657Thrfs*19). Next-generation sequencing (NEBNext kit, New England Biolabs) showed the same *NF1* CH finding (VAF, 5.2%). The patient did not take any standing medications. On examination, the heart rate was 70 bpm, and the blood pressure was 150/90 mmHg, with a body mass index of 32.1 kg/m². Laboratory examinations showed a normal hemogram, total cholesterol of 140 mg/dL, triglycerides of 117 mg/dL, LDL-C of 76 mg/dL, HDL-C of 41 mg/dL, and HgA1C of 5.2%.

Patients with cancer often undergo genomic profiling via DNA sequencing of neoplastic tissue and peripheral blood to detect tumor-specific genomic aberrations with the goal of establishing a personalized molecular target for therapy. Although such assays aim to identify pathogenic germline variants, this process may also disclose acquired somatic mutations. The finding of somatic mutations may provide important insights into aging and cancer development by revealing how specific clones expand over time and drive tissue dysfunction and malignant transformation. Differentiating between acquired and germline variants often informs management decisions and disease prognostication.

Among patients with solid tumors who undergo germline testing, CH is found in approximately one-third of

the cases.¹¹ Similar to the overall distribution of mutations in individuals with CHIP, *DNMT3A*, *TET2*, *PPM1D*, and *ASXL1* are the most prevalent mutated genes.^{11,12} Congruent with data derived from prospective biobanks, older age, smoking history, and genetic ancestry are all associated with the presence of CH. Interestingly, an association between pathogenic and likely pathogenic germline mutations and the likelihood of CH has been demonstrated. Specifically, germline mutations in the *APC*, *CHEK2*, and *ATM* genes show the highest odds of harboring CH, an association that remains significant after covariate adjustment in the case of *ATM* (odds ratio, 1.58).¹² Moreover, tumor tissue may harbor cells bearing CHIP mutations, a finding that associates with worse prognosis.

Specific CH genotypes are more prevalent among patients with different germline cancer-predisposing mutations. For example, *DNMT3A* CH occurs more often in patients with germline *ATM* mutations. In addition, *TET2* CH prevalence is greater among patients with germline *FANCA* mutations. On the other hand, *CHEK2* demonstrates a negative trend for the presence of CH although an analysis of putative loss-of-function germline variants has shown a modest association.^{11,13} Other pathogenic inherited loci involving *TERT*, *SH2B3*, and *TET2* have been associated with development of age-related *JAK2* V617F CH and development of myeloproliferative neoplasms independent of *JAK2* V617F.¹⁴ Furthermore, in individuals with mosaic chromosomal alterations, rates of CH approximate to 50%, and an intriguing association has emerged for positive selection for *CHEK2* biallelic inactivation and mosaic chromosomal alterations of loss-of-heterozygosity type. These associations highlight the potential biologic function of somatic mutations and mosaic chromosomal alterations in the biallelic inactivation of DNA damage response (DDR) genes.

Case Continued

We counseled the patient that *NF1* CH is rare, and its cardiovascular implications remain unknown because existing CHIP-cardiovascular disease association studies have insufficient gene-specific power to evaluate *NF1*. We recommended losartan 25 mg daily to achieve better blood pressure control and counseled her on the importance of lifestyle measures to lower atherogenic lipid particles.

CH IN PATIENTS WITH EXISTING MALIGNANCY

Case 3

A 53-year-old female with polycythemia vera, familial hypercholesterolemia, transient ischemic attack, past cigarette smoking, and family history of premature coronary

artery disease was referred to our clinic for cardiovascular assessment and management. Her polycythemia vera was characterized by a *JAK2* exon 12 mutation, and she was noted to have a *DNMT3A* mutation (VAF, 1.3%). She had received cytoreduction with hydroxyurea for several years. The patient reported no symptoms at rest or with exertion. Her regular medications otherwise included rosuvastatin 20 mg daily, ezetimibe 10 mg daily, and aspirin 81 mg daily. On examination, the heart rate was 66 bpm, and the blood pressure was 118/76 mm Hg, with a body mass index of 34.4 kg/m². On cardiac auscultation, a faint mid-systolic murmur was appreciated in the upper sternal border territory. Laboratory investigations showed a normal hemogram, total cholesterol of 175 mg/dL, triglycerides of 97 mg/dL, LDL-C of 91 mg/dL (previously >200 mg/dL), HDL-C of 65 mg/dL, Lp(a) (lipoprotein [a]) of 515 nmol/L, and HgA1C of 5.0%.

CH can accompany various human cancers, as demonstrated by comprehensive studies of cancer genomes.¹⁵ The frequency of CH, either in the form of somatic point mutations or mosaic chromosomal alterations, varies widely within a given cancer type and between different types of cancers. This variability reflects both intrinsic tissue features (eg, cellular architecture and turnover) and exposure to endogenous and exogenous mutagenic stressors, which together influence the background rate and spectrum of somatic alterations observed across tissues.¹⁵ For example, cancers that develop in tissues with high cellular turnover and with exposure to exogenous mutagens, such as lung, gastric, and colorectal cancer, tend to have a high prevalence of base substitutions.¹⁶ On the other hand, testicular cancers and breast malignancies have a significantly lower frequency of mutations. Aside from tissue origin, prior treatment with cytotoxic drugs and defective DNA damage and repair leading to microsatellite instability also correlate with the likelihood of somatic mutations.^{16,17}

Myeloproliferative neoplasms can arise from somatic mutations in the *TET2*, *IDH1*, *IDH2*, *DNMT3A*, and *CALR* genes.¹⁸ In a study of patients with essential thrombocythemia or primary myelofibrosis, analysis of hematopoietic-progenitor colonies determined that somatic mutations in *CALR* were acquired early and associated with higher platelet count and lower hemoglobin count and predicted longer overall survival compared with those with *MPL* or *JAK2* mutations in myelofibrosis but not in essential thrombocythemia.¹⁹ Patients with *JAK2*-positive myeloproliferative neoplasms can also have mutations involving the *TET2* loci. Although, in some cases, *TET2* loss was homozygous, potentially invoking its tumor suppressor activity in the development of mutations, the low percentage of homogenous colonies implied that *TET2* was considered unlikely to confer a competitive advantage.²⁰ Furthermore, the lack of a strict temporal relationship between *TET2* mutation emergence and the acquisition of *JAK2* mutation has led

investigators to speculate that the former does not meaningfully contribute to predisposition to acquire *JAK2* or to the development of myeloproliferative neoplasms.²⁰ The impact of coexisting somatic mutations in *TET2*, or point mutations in other myeloid family genes, on the risk of cardiovascular disease deserves additional study.

Case Continued

We discussed that this patient's cardiovascular risk is predominantly driven by *JAK2*-mutant polycythemia vera, markedly elevated Lp(a), and obesity, while the low-VAF *DNMT3A* mutation likely contributes little or no incremental risk. We recommended intensification of strategies to control modifiable risk factors, including her LDL-C, with the addition of Proprotein Convertase Subtilisin/Kexin Type 9 inhibitor to her statin and ezetimibe, with a goal of LDL-C <70 mg/dL. Given the patient's mid-systolic murmur and its known association with aortic stenosis, we ordered a transthoracic echocardiography. We also discussed cascade screening of her siblings.

CH FOLLOWING CYTOTOXIC ANTICANCER THERAPY

Case 4

An 82-year-old man with a medical history of hypertension, chronic kidney disease, radiographic coronary artery disease, and nonsmall cell lung cancer status post-immunotherapy and thoracic irradiation was referred to us for CH. Cytopenia detected during surveillance, 4 years after completion of definitive therapy for his thoracic malignancy, prompted next-generation sequencing (NEBNext kit, New England Biolabs), leading to the finding of *PPM1D* CH (VAF, 4.8%). The patient's regular medications included metoprolol tartrate 12.5 mg twice daily. On examination, the heart rate was 65 bpm, and the blood pressure was 125/72 mm Hg, with a body mass index of 26.8 kg/m². Ankle brachial index was 1.1 (right) and 1.2 (left). Laboratory examinations showed total cholesterol of 160 mg/dL, triglycerides of 149 mg/dL, LDL-C of 90 mg/dL, HDL-C of 40 mg/dL, Lp(a) of 28 nmol/L, and HgA1C of 5.7%.

Exposure to cytotoxic therapies, such as chemotherapy and radiation therapy, is closely associated with higher odds of acquiring CH. In particular, platinum and topoisomerase II inhibitors favor the selection of mutations in DDR genes, such as *TP53*, *PPM1D*, and *CHEK2*.²¹ Expansion of these clones tends to be faster compared with non-DDR clones. Conversely, non-DDR CH tends to grow faster than DDR genes in cohorts of patients who have not been exposed to mutagenic agents.²¹

In a prospective study of women with breast cancer, the rate of new CH following chemotherapy did

not differ statistically from that of patients treated with endocrine therapy.²² However, new CH with VAF >0.005 was 3× more frequent in patients treated with chemotherapy.²² Furthermore, cytotoxic exposure was associated with an elevated risk of developing a new high-risk clone involving *TP53* (odds ratio, 3.57), in as many as 4% of the patients. In a prospective study of patients with esophageal or lung cancer who had sequencing data before and after chemoradiation, the number of *TP53* mutations doubled. In addition, expansion of clone size was noted in 38% of patients and a decrease in clone size in 5% of the cases. An increase in the number of *TP53* mutations post-chemoradiation was associated with a substantially greater risk of mortality (hazard ratio, 7.1).²³

PPM1D encodes a stress-response phosphatase that mutes the DNA damage response, and *PPM1D*-mutant hematopoietic clones occur preferentially after cytotoxic exposures. Epidemiologically, *PPM1D*-CH has been associated with increased atherosclerotic cardiovascular disease risk, with signals suggesting enrichment for disease in peripheral arterial beds in some cohorts, consistent with a proinflammatory myeloid phenotype that may enhance myeloid cell proliferation in the plaque and boost its progression.²⁴ Beyond atherosclerosis, emerging population-scale analyses have also begun to link CH, including *PPM1D*-mutant CH, with incident atrial fibrillation, supporting the concept that clonal, inflammation-skewed leukocyte populations may contribute not only to vascular disease but also to atrial remodeling and electrical instability.

Case Continued

We counseled the patient on the association between *PPM1D* CH and atherosclerotic cardiovascular disease. We recommended the initiation of atorvastatin 40 mg daily, targeting LDL-C <55 mg/dL in the setting of a high-risk mutation in the background of coronary artery disease and prior thoracic irradiation. We discussed the importance of nicotine avoidance (including second-hand) and regular aerobic activity.

SUMMARY

CH may be detected in a variety of clinical settings in otherwise healthy individuals and in cancer survivors (Figure). Although no consensus guidelines exist for routine longitudinal surveillance of CH, repeat sequencing at 12- to 24-month intervals is commonly used in higher-risk settings such as CCUS, postcytotoxic exposure, or worsening blood count abnormalities, while routine retesting is not otherwise generally performed. The tempo of retesting could consider the specific mutation, as *TET2* clones tend to expand rapidly, while those with *DNMT3A* mutations expand more slowly.

That said, the rate of clone growth does not currently alter management due to data dearth. Our general approach to the cardiovascular prevention of individuals with CH centers on the evaluation of high-risk features, which include the presence of high-risk mutations (*TET2*, *JAK2 V617F*, and *TP53*), large VAF (>10%), >1 mutations, and traditional cardiovascular risk factors. However, delineation of the context in which the CH is identified is important when assessing the biological and clinical implications of the clonal process. For example, *TET2* CH in patients with concomitant hematologic malignancy has a predilection for significant expansion over time and, thus, greater cumulative risks of hematologic malignancy and ischemic cardiovascular events. Somatic mutations emerging following cytotoxic therapy, although typically showing clonal stability after 1 to 2 years from detection, tend to involve DDR genes, which confer increased mortality rates in the case of *TP53* and accelerated multiarterial atherosclerosis, with a particular predilection for peripheral artery disease. On the other extreme are elderly individuals with CH involving nonmyeloid genes with VAF<2%, in whom the possibility of age-related CH should be entertained, for which associations with future cardiovascular disease risk are unknown. In all cases, and particularly when germline testing reveals CH, the question of somatic mosaicism must be addressed. To address that, cardiologists must work in concert with the patient's hematologist, incorporating the use of germline testing, longitudinal clonal size assessment, and obtaining a family history, to distinguish somatic mosaicism from true CH. Any assessment of the impact of CH on an individual's health should not only factor in the somatic mutation and its size but also past and current oncological and cardiovascular medical history, tumor type, and its associated prognosis, as well as the context in which testing was performed.

Classification of individuals with CH cases into defined categories, as presented here, also has value for the biologically informed design and conduct of clinical trials in the field of CH. As demonstrated above, mutation kinetics tend to differ in the various settings and possible therapeutic interventions to suppress clonal activity, and consequent events should be tailored not only to the specific aberrant gene mutation, for example, the associated activated inflammasome, but also to the typical natural history of such abnormal clones. For instance, inflammasome inhibition might be more beneficial in hyper-inflammatory patients with active malignancies and those receiving proinflammatory therapies such as large fields of radiation or immune checkpoint inhibitors, while mutation-specific inhibitors, such as *JAK2* inhibitors, might preferentially ameliorate the prothrombotic tendency in patients with coronary artery plaques who harbor a large burden of CH. Randomization of well-defined cohorts, characterized not only by

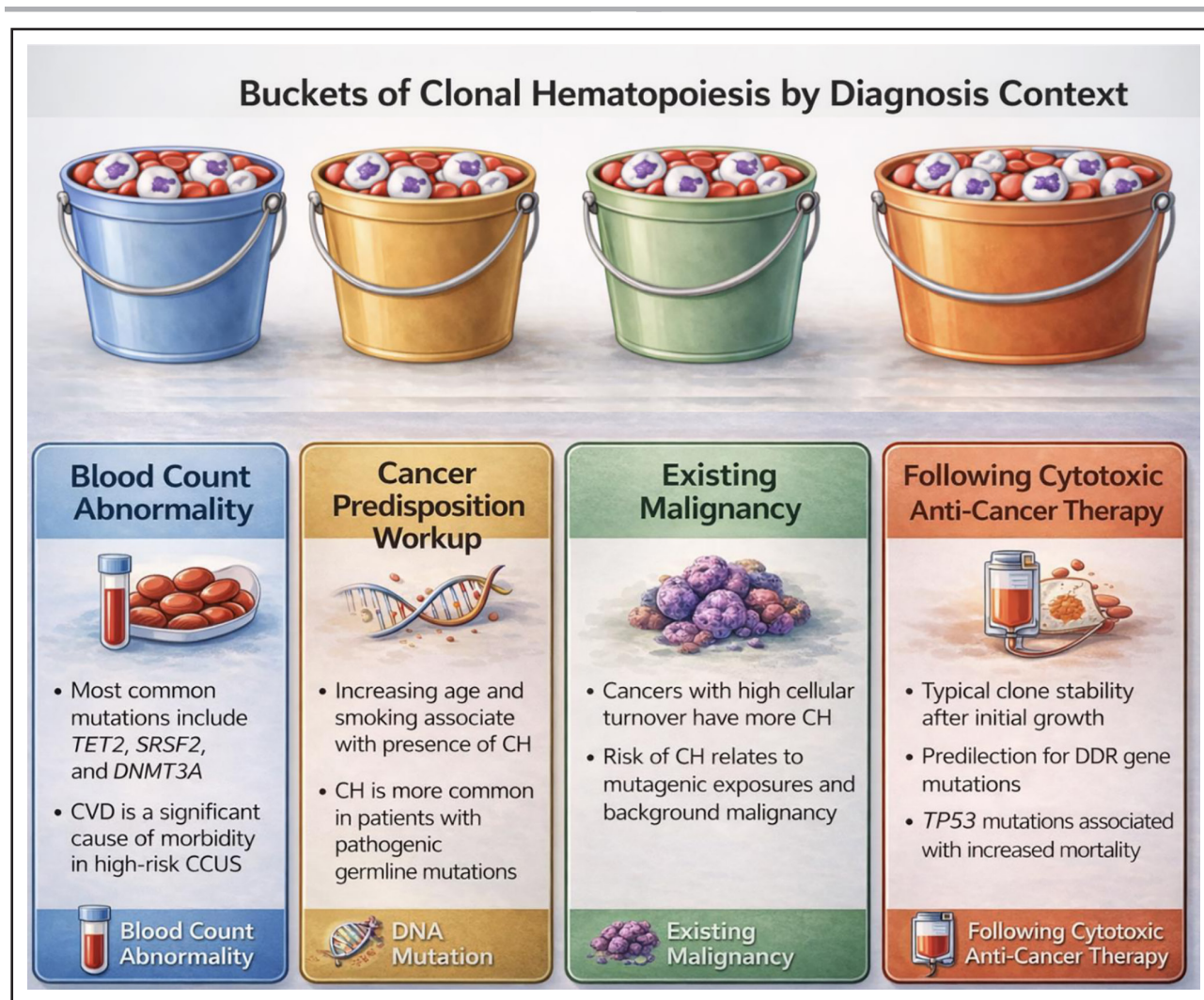


Figure. Spectrum of patients referred to the cardiovascular specialist with the question of clonal hematopoiesis (CH).

Detection of CH may occur in several distinct settings representing different triggers for diagnostic sequencing and different baseline health states and exposure to mutagenic therapies. CCUS indicates clonal cytopenia of unknown significance; CVD, cardiovascular disease; and DDR, DNA damage response.

their hallmark mutation and clone size but also by the diagnostic context in which CH was identified, has the potential to enhance the biological rationale and efficacy of costly, long-term clinical trials in this field.

ARTICLE INFORMATION

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